

Distributional Dynamic Programming in Economics

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Abstract

This paper introduces a new family of decision-making models for agents in structural economic environments, emphasizing the consideration of the entire return distribution rather than just the expected return. We extend the traditional framework of expectation maximization by incorporating various measures over the distribution function, allowing for a more nuanced analysis of agent behavior under uncertainty.

The proposed framework, termed *Distributional Dynamic Programming (DDP)*, generalizes conventional dynamic programming to focus on the *value distribution function*, from which optimal policies are derived. We develop the *Distributional Value Function Iteration (DVFI)* method, which iteratively applies the *Distributional Bellman Operator* and *M*-quantile projections to solve for the value distribution function.

We apply this framework to two seminal models in economics: the McCall Search Model in labor economics and the Aiyagari Model in macroeconomics. These applications demonstrate how risk aversion, concern for specific quantiles, and different preferences over the distribution function can significantly influence decision-making processes and equilibrium outcomes.

Our findings suggest that incorporating distributional considerations into dynamic programming models offers valuable insights and provides a more comprehensive understanding of decision-making under uncertainty. This approach opens up new avenues for research, particularly in exploring the implications of alternative distributional measures in economic modeling.

Keywords:

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1 Introduction

This paper contributes to the literature by proposing a family of alternative decision-making models for agents in structural environments. In our framework, agents consider not the expected return, but the entire return distribution. Each decision model is characterized by the maximization of a specific measure of the distribution function. This approach encompasses the traditional expectation maximization framework as a special case when the measure corresponds to the first moment. By doing so, we generalize the discussion and introduce a new dimension to the analysis.

The economics literature has traditionally viewed agents as expectation maximizers in dynamic decision-making (see, for example, Samuelson [1983], Koopmans [1960], Brock and Mirman [1972]). This approach offers a straightforward method to compare different options and outcomes. However, substantial evidence indicates that human decision making is more nuanced, particularly under uncertainty [de Castro et al., 2022]. Having individuals care only about the first moment of variation may not fully capture the complexity of their behavior.

Consider a scenario where you need to catch an early morning flight for a conference and want the shortest trip to the airport to maximize your sleep. Routine A consistently takes 1.5 hours, while Routine B usually takes 1.2 hours but has a 10% chance of requiring maintenance, potentially extending the trip to over 3 hours, causing you to miss the flight. An expectation maximizer would likely choose Routine B for its shorter average time, but someone who places significant weight on the risk of missing the flight might prefer Routine A.

The traditional approach to incorporating risk aversion involves adopting a utility function over the hours taken, with a risk aversion parameter that adjusts the curvature of the function. In this way, the expected utility, rather than the hours alone, reflects the agent's risk preferences. However, defining such a utility function for a particular problem is not always feasible or straightforward. An alternative approach, as proposed in this paper, is to directly model the distribution of the hours and encode the agent's preferences through a measure over this distribution that balances the trade-off between expectation and variance.

More specifically, this paper considers the Markov decision-making process of an agent, as is common in many dynamic economic models. We introduce *Distributional Dynamic Programming (DDP)* by generalizing the discussion in conventional dynamic programming, which typically focuses on the value function, to consider the so-called *value distribution function* of the agent. Analogous to the case where an agent derives the optimal policy from the value function, the optimal policy is now derived from the value distribution function, with respect to a pre-chosen measure that maps the value distribution function to a scalar value. We propose various measures that reflect different preferences over the distribution.

We then discuss the properties of the value distribution function, showing that it is the fixed point of the *distributional Bellman operator*, just as the value function is the fixed point

of the conventional Bellman operator. We consider the p -Wasserstein distance as a metric in the function space, noting that the distributional Bellman operator has been proven to be a contraction mapping with a convergence rate equal to the discount rate. Hence, we can obtain the fixed point by iteratively applying the operator to an initial guess.

However, for implementation, we must address the practical issue of how to characterize a distribution. In this paper, we consider the M -quantile parameterization, which characterizes a distribution by M equally weighted particles. We then introduce a projection method that ensures that the projected distribution has the smallest 1-Wasserstein distance to the original distribution, thus achieving a w_1 -projection.

Finally, we combine all these components and propose an algorithm to implement the distributional Bellman operator, alternating with the M -projection, until the value distribution function converges and the policy function is obtained. We call this approach *Distributional Value Function Iteration (DVFI)*, an analogy to Value Function Iteration (VFI) in the context of handling the value function.

For applications, we consider two foundational models in the economics literature: the McCall Search Model [McCall, 1970] in labor economics and the Aiyagari model [Aiyagari, 1994] in macroeconomics. In the former, we extend the agent’s decision-making process from merely trading off between accepting and rejecting a randomly drawn offer based on expected cumulative discounted income to scenarios where the agent is risk-averse. This includes cases where the agent imposes varying degrees of penalty on income variance if the current offer is rejected, as well as situations where the agent considers specific quantile levels of the return. In the latter application, where general equilibrium is characterized by the wealth distribution among agents, we examine whether the Distributional Value Function Iteration (DVFI) produces the same solution as the Value Function Iteration (VFI) when the first moment is chosen as the measure of agents’ preferences over the distribution. In future work, we will explore how different preferences, determined by the choice of measure in the value distribution function, impact the stationary distribution of the economy. This will be compared to the benchmark case where the agent is an expectation maximizer.

The remainder of the paper is structured as follows: Section 2 provides the literature review; Section 3 introduces the technical aspects of the Distributional Dynamic Programming approaches; Section 4 presents the first computational example, the McCall Search Model, and Section 5 discusses the second example, the Aiyagari model. In Section 6, we briefly outline various directions for future work, and finally, Section 7 concludes the paper.

2 Literature Review

This paper is closely related to the work of de Castro and Galvao [2019], which developed a dynamic model where agents maximize the stream of future τ -quantile utilities. In their

subsequent work [de Castro et al. \[2023\]](#), the authors studied numerical methods for the quantile preference model, and in [de Castro et al. \[2022\]](#), they provided empirical evidence supporting agents’ quantile maximization behavior. Our work generalizes these results by allowing agents to maximize a particular measure of the distribution of returns. The two approaches coincide when the chosen measure is a quantile measure.

This paper builds on ideas from a rapidly growing body of literature on distributional approaches in reinforcement learning, as seen in works such as [Bellemare et al. \[2017\]](#), [Dabney et al. \[2018b\]](#), [Dabney et al. \[2018a\]](#), and [Lyle et al. \[2019\]](#). Notably, many of the technical remarks in [Section 3](#) are drawn from the recent survey by [Bellemare et al. \[2023\]](#). The main contributions of this paper are twofold: first, we apply distributional approaches within economic environments, which are inherently stochastic with various sources of uncertainty. The discussion of risk-averse agents in economics dates back several decades, highlighting the necessity of distributional decision-making models in this context. Second, we explore distributional decision-making within the dynamic programming framework, which is prevalent in economics where agents are typically assumed to have knowledge of the state transition dynamics. This contrasts with the deep reinforcement learning literature mentioned above, where agents often learn the environment’s transitions through trial and error, either in real-world experiments or simulators.

3 Technical Remarks

3.1 From Value Function to Value Distribution Function

To begin with, we focus on the discrete-time Markov decision process (MDP), which is a tuple $(\mathcal{S}, \mathcal{A}, \gamma, P_{\mathcal{S}}, P_{\mathcal{R}})$, where $\mathcal{S}$ is the state space: state $S_t \in \mathcal{S}$, $\mathcal{A}$ is the action space: action $A_t \in \mathcal{A}$, γ is the discount rate, and $P_{\mathcal{S}}$ and $P_{\mathcal{R}}$ are the transition kernel and reward distribution functions, respectively. Specifically, $R_t \sim P_{\mathcal{R}}(\cdot \mid S_t, A_t)$ and $S_{t+1} \sim P_{\mathcal{S}}(\cdot \mid S_t, A_t)$, where we assume that the reward R_t and the next state S_{t+1} are independent given S_t and A_t . Since the reward and transition are conditional on the action, the description of the MDP is complete given a specific policy, which is a mapping $\pi : \mathcal{S} \rightarrow \mathbb{P}(\mathcal{A})$ from states to probability distributions over actions such that $A_t \sim \pi(\cdot \mid S_t)$.

We are interested in the accumulated discounted return of an agent obtained by following a particular policy π from a given state s in an MDP:

$$G^{\pi}(s) = \sum_{t=0}^{\infty} \gamma^t R_t, \quad S = s$$

Note that since $\{R_t\}_{t=0}^{\infty}$ is a sequence of random variables, their sum G^{π} is itself a random variable. The bounded reward $R_t \in [R_{\min}, R_{\max}]$ implies $G^{\pi} \in [\frac{R_{\min}}{1-\gamma}, \frac{R_{\max}}{1-\gamma}]$. One of the

typical interest is the expectation of the random variable G^π , and accordingly the optimal policy π^* that maximize it, which is the familiar state value function:

$$V^*(s) = \mathbb{E}_\pi[\max_\pi \sum_{t=0}^{\infty} \gamma^t R_t | S = s] = \mathbb{E}_\pi[G^*(s)] \quad (1)$$

We use the superscript $*$ to denote any function or variable associated with the optimal policy π^* . Another function of particular interest, especially in the computer science literature, is the state-action value function, commonly known as the “ Q -function” (Watkins and Dayan [1992], see also Sutton [2018]):

$$Q^*(s, a) = \mathbb{E}_\pi[\max_\pi \sum_{t=0}^{\infty} \gamma^t R_t | S = s, A = a] = \mathbb{E}_\pi[G^*(s, a)] \quad (2)$$

where, with a slight abuse of notation, we use G to now denote $G^*(s, a)$, the accumulated discounted return obtained by following the optimal policy π^* from a given state s and action a , and the Q -function represents its expectation. The Q -function has some computational convenience since we can easily recover V^* and obtain the optimal policy function π^* from Q^* :

$$V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a) \quad (3a)$$

$$\pi^*(s) = \arg \max_{a \in \mathcal{A}} Q^*(s, a) \quad (3b)$$

Therefore, from this point forward, we will focus our discussion on the Q -function. Whenever we refer to the “value function,” we specifically refer to the state action value function, the Q -function. However, note that all the results presented have an analogous version for the standard value function V .

While much of the literature focuses on the expectation of random returns, this paper broadens the discussion to include any measure of the random variable G^* , or equivalently, its distribution. Specifically, we introduce the notation η^* , known as the *value distribution function*, where $\eta^*(s, a)$ represents the distribution of $G^*(s, a)$ for a given state-action pair (s, a) .

Next, we introduce the measure function $\rho : \mathcal{P}_\rho(\mathbb{R}) \rightarrow [-\infty, \infty)$, which maps a distribution function to a scalar. This measure function characterizes the agent’s specific preference over distributions, allowing us to compare and rank different distributions based on the chosen measure.

Our objective is to seek an optimal policy π^* such that for each state s we maximize a

generalized value of return measure by a particular ρ :

$$\pi^*(s) = \arg \max_a \rho(\eta^*(s, a)) \quad (4)$$

That is, we derive the optimal policy function π^* once we have the value distribution function η^* . In the next subsection we describe how to compute η^* . Below, we consider some particular families of measures:

- Measures over moments of the distribution: Let the moments of a distribution up to order L be denoted as $m = (m_1, \dots, m_L)$. A measure $\rho_{\text{moment}} = h(m)$ can take a specific functional form that reflects the agent's trade-off between different moments of interest. For example, the mean: $\rho_M(\eta) = \mathbb{E}(\eta) = m_1$, which corresponds to the standard value function; or the mean-variance trade-off, as commonly discussed in the finance literature: Let $\lambda > 0$ be the penalty parameter on variance, then $\rho_{MV}^\lambda(\eta) = \mathbb{E}(\eta) - \lambda\sqrt{\mathbb{V}(\eta)} = m_1 - \lambda\sqrt{m_2}$.
- Measures over specific quantile levels/Value-at-Risk: Let $\tau > 0$. The measure ρ_Q^τ represents the τ -quantile of the distribution, defined as $\rho_Q^\tau(\eta) = F_\eta^{-1}(\tau)$.
- Measures based on entropy: denote $f_\eta(x)$ the PDF of η , the entropy measure ρ_H is defined as $\rho_H(\eta) = -\int f_\eta(x) \log f_\eta(x) dx$. This measure captures the degree of uncertainty or dispersion in the distribution.
- Measures over the distance from a benchmark: Let $\hat{\eta}$ be a benchmark distribution that the agent prefers η to closely resemble, according to a metric in function space $g : \mathcal{P}(\mathbb{R}) \times \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$. The measure $\rho_{\hat{\eta}}^g$ is then defined as $\rho_{\hat{\eta}}^g = g(\eta, \hat{\eta})$.

3.2 Distributional Bellman Operator

To write down the recursive form of an MDP problem, we drop the time subscript t and use X' to denote the variable X in the next period. It is well known that, under mild conditions, Q^* is the unique fixed point of the celebrated Bellman operator (Bellman [1966]):

$$\begin{aligned} (\mathcal{T}Q)(s, a) &= r(s, a) + \gamma \int_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} Q(s', a') f(s' | s, a) ds', \quad s \in \mathcal{S}, a \in \mathcal{A} \\ Q^* &= \mathcal{T}Q^* \end{aligned} \quad (5)$$

Given the current estimated Q -value of a particular (s, a) pair, the Bellman operator consists of three operations to obtain the new estimate:

- Maximization: conditional on s' , we choose the a' that gives us the maximum value.

- Integration: we integrate over s' governed by the transition function with (s, a) .
- Scaling: We shift the integrated value by defining the bootstrap function $b_{r,\gamma} : \mathbb{R} \rightarrow \mathbb{R}$, indexing by the reward $r \in \mathbb{R}$ and discount $\gamma \in [0, 1)$, such that $b_{r,\gamma}(z) = r + \gamma z$.

Intuitively, being the fixed point of the Bellman operator is the constraint imposed on Q^* , the first moment of the random variable G^* . To generalize this idea, an extended version of the Bellman operator, the so-called *Distributional Bellman Operator* (Bellemare et al. [2017]) has been proposed to characterize η^* , the whole distribution of G^* :

$$(\mathcal{T}\eta)(s, a) = \mathcal{B}_{r,\gamma} \int_{s' \in \mathcal{S}} \eta(s', a') f(s' | s, a) ds' \quad (6a)$$

$$\text{where } a' = \arg \max_{a' \in \mathcal{A}} \rho(\eta^*(s', a')) \quad (6b)$$

$$\eta^* = \mathcal{T}\eta^* \quad (6c)$$

where we use the same symbol $\mathcal{T}$ for the operator for simplicity of notation. Analogous to the case of the benchmark Bellman operator, we can understand the distributional Bellman operator as a composition of three operations:

- Maximization over ρ : conditional on s' , we choose a' that gives us the maximum value of the function under the measure ρ .
- Integration: the summation above becomes the convolution over different distributions η , which is technically more challenging and we will discuss this in more detail in the next subsection.
- Scaling: Similarly to the case above, but now $\mathcal{B}_{r,\gamma}$ is a bootstrap operator in the integrated distribution, which means that we apply the mapping $z \rightarrow r + \gamma z$ on every realization of η .

We say that η^* satisfies *the distributional Bellman equation* if it is the solution to 6, analogous to how Q^* is the solution to the Bellman equation. However, this does not provide a method for obtaining such a solution. In the case of Q^* , we know that it can be found by iteratively applying the Bellman operator to an initial guess Q until convergence, since the Bellman operator is a contraction mapping in vector space. To generalize this result and solve for η^* by iteratively applying the distributional Bellman operator to an initial guess η , two conditions must be met: (1) a defined metric space for the value distribution function η and (2) proof that the distributional Bellman operator is a contraction mapping in this function space. In the following, we briefly summarize results from the current literature, focusing on one of the metrics commonly used in the space of distribution functions, the p -Wasserstein distance.

Definition 1 (Kantorovich [1960]). Let $\eta \in \mathcal{P}(\mathbb{R})$ be a distribution function with its CDF denoted as F_η , the inverse F_η^{-1} is given by $F_\eta^{-1}(\tau) = \inf_{z \in \mathbb{R}} \{z : F_\eta(z) \geq \tau\}$. Let $p \in [1, \infty)$. The p -Wasserstein (w^p) distance is a function $g_p : \mathcal{P}(\mathbb{R}) \times \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ given by

$$g_p(\eta, \eta') = \left(\int_0^1 |F_\eta^{-1}(\tau) - F_{\eta'}^{-1}(\tau)|^p d\tau \right)^{1/p}.$$

Graphically, the Wasserstein distances between two probability distributions measure the area between their CDF, with values along the abscissa taken to the p -th power. In terms of the value distribution function η , we consider the supremum p -Wasserstein distance over the (s, a) pairs:

Definition 2. Let $p \in [1, \infty]$. The supremum p -Wasserstein distance $\bar{g}_p$ between two value distribution functions $\eta, \eta' \in \mathcal{P}(\mathbb{R})^{S \times \mathcal{A}}$ is defined by

$$\bar{g}_p(\eta, \eta') = \sup_{s \in S, a \in \mathcal{A}} g_p(\eta(s, a), \eta'(s, a)).$$

Based on the supremum p -Wasserstein distance, it has been shown that the distributional Bellman operator is contractive, as follows:

Proposition 1. The distributional Bellman operator is a contraction mapping on $\mathcal{P}(\mathbb{R})^{S \times \mathcal{A}}$ in the supremum p -Wasserstein distance for all $p \in [1, \infty]$. More precisely,

$$\bar{g}_p(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \eta') \leq \gamma \bar{g}_p(\eta, \eta'),$$

for all $\eta, \eta' \in \mathcal{P}(\mathbb{R})^{S \times \mathcal{A}}$.

The proof can be found in Chapter 4 of Bellemare et al. [2023]. This proposition demonstrates that iteratively applying the distributional Bellman operator to an initial guess of the value distribution function η_0 guarantees convergence. However, two practical challenges remain:

- Characterizing a distribution function on a computer.
- Ensuring that the chosen characterization is closed under the distributional Bellman operator, which requires applying a projection to ensure that the new estimate belongs to the same family of characterizations.

These issues are addressed in the next subsection.

3.3 Parameterization and Projection

Denote $\mathcal{F}$ a particular parameterization. To approximate an arbitrary distribution $\eta \in \mathcal{P}(\mathbb{R})$ by $\mathcal{F}$, we introduce the projection operator $\mathcal{P} : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{F}$. Intuitively, $\mathcal{P}$ chooses a distribution $\hat{\eta}$ from $\mathcal{F}$ to minimize the difference from η measured by a given metric d :

$$\mathcal{P}(\eta) = \arg \max_{\hat{\eta} \in \mathcal{F}} d(\eta, \hat{\eta}) \quad (7)$$

In this case, we call $\mathcal{P}$ a d -projection. Here, we consider minimizing the 1-Wasserstein (w^1) distance and one of the parameterizations of the distribution function that is suitable to this metric, the M -quantile parameterization.

Definition 3. *The M -quantile parameterization of a distribution function η is given by the set:*

$$\mathcal{P}_Q^M(\eta) \in \mathcal{F}_Q^M = \left\{ \frac{1}{M} \sum_{m=1}^M \delta_{\theta_m} : \theta_m \in \mathbb{R} \right\} \quad (8)$$

That is, the M parameters $\{\theta_m\}_{m=1}^M$ characterizes the location of M equally weighted particles. The following proposition gives the optimal selection of θ when we choose the 1-Wasserstein distance:

Proposition 2. *Let $\eta \in \mathcal{P}(\mathbb{R})$. The M -quantile probability distribution whose parameters are given by*

$$\theta_i = F_\eta^{-1} \left(\frac{2m-1}{2M} \right) \quad m = 1, \dots, M \quad (9)$$

is a w_1 -projection of η onto $\mathcal{F}_{Q,M}$. We denote as $\tau_M = \{\frac{2m-1}{2M}\}_{m=1}^M$ the quantile levels determined by the choice of M .

The proposition states that each interval $[\frac{m-1}{M}, \frac{m}{M}]$, $m = 1, \dots, M$ is characterized by the midpoint $\frac{2m-1}{2M}$ quantile. Figure 1 illustrates the case where we approximate a random distribution η by the step-wise linear function $\hat{\eta}$ choosing $M = 5$.

The M -quantile parameterization requires knowledge of the exact values associated with τ_M . On the other hand, there is an integration step in the distributional Bellman operator over the integrands indexed by the next state s' and weighted by the transition function $f(s'|s, a)$. By applying the distributional Bellman operator in each iteration, even if the integrands are M -quantile projected (which can be achieved by initialization), the integral is not guaranteed to be M -quantile projected. In other words, the M -quantile projection is not guaranteed to be closed under the distributional Bellman operator.

We use an example to demonstrate the implementation of the integration and subsequent projection of a set of M -quantile projected integrands. Again, we choose $M = 5$ so that $\tau_M = (0.1, 0.3, 0.5, 0.7, 0.9)$. There are two distributions $\eta_1, \eta_2 \in \mathcal{F}_{Q,M}$ such that η_1 is

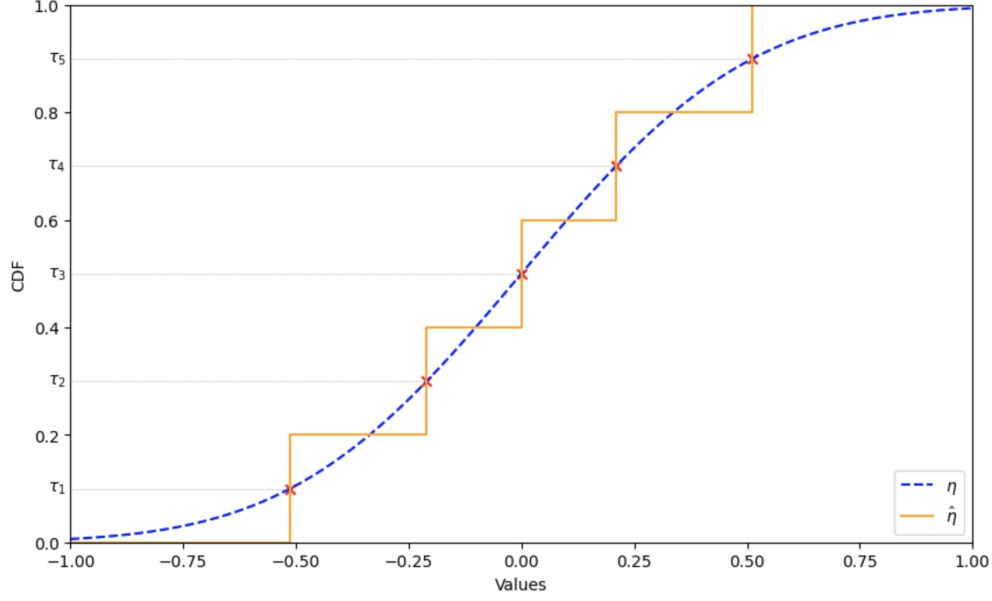


Figure 1: An example of quantile projection. By choosing $M = 5$, we divide the support into the intervals $([0, 0.2], [0.2, 0.4], [0.4, 0.6], [0.6, 0.8], [0.8, 1.0])$ and the associated midpoints are $(0.1, 0.3, 0.5, 0.7, 0.9)$.

characterized by $X_1 = (1, 3, 5, 7, 9)$ and η_2 is characterized by $X_2 = (0, 2, 6, 10, 10)$. η_3 is a mixture of η_1 and η_2 with weights $W = (0.4, 0.6)$: $\eta_3 = W_1\eta_1 + W_2\eta_2$. Our goal is the M -quantile projection of η_3 .

Since each function is characterized by equally weighted particles (in this case, $\frac{1}{M} = 0.2$ for each), the mixture distribution has each particle with a new weight equal to the original weight times the weight in W . Then, by ordering the particles according to their values, we can find the corresponding value for each quantile level in τ_M . Specifically, we first sort the combined set of values $X_3 = (X_1, X_2)$, then order the associated weights for each particle: for example, the mass of each particle in η_2 is $W_2/M = 0.12$; and finally we search for the value corresponding to each quantile level in τ_M in the new distribution. Figure 2 gives an illustration of this example.

We generalize the implementation for the case of K value distribution functions $\{\eta_k\}_{k=1}^K \in \mathcal{F}_Q^M$, where $K = |\mathcal{S}|$ is the size of the state space, as stated in Algorithm 1 below.

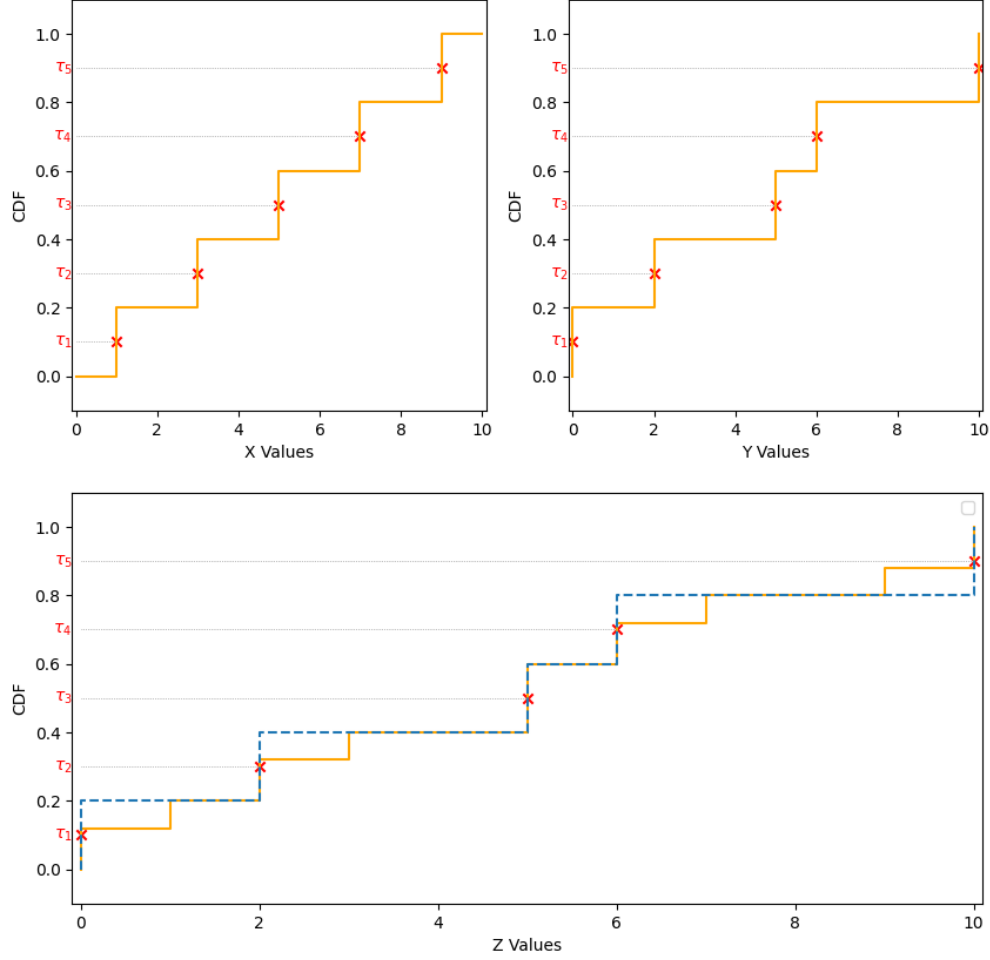


Figure 2: Illustration of the integration and subsequent projection of M -quantile projected integrands. $M = 5$ and $\tau_M = (0.1, 0.3, 0.5, 0.7, 0.9)$. Upper left: η_1 with values $X_1 = (1, 3, 5, 7, 9)$. Upper right: η_2 with values $X_2 = (0, 2, 6, 10, 10)$. Bottom: blue solid line: the integral η_3 before projection; orange dashed line: the integral η_3 after projection.

Algorithm 1 Integration and Projection of M -Quantile Projected Integrands

- 1: **Parameters:** K : number of integrand functions, M : number of quantile levels to characterize the distributions
 - 2: **Input:** M -quantile levels $\tau_M = \{\frac{2m-1}{2M}\}_{m=1}^M$, Distributions $\{\eta_k\}_{k=1}^K \in \mathcal{F}_Q^M$ characterized by $\{X_k = (x_{mk})_{m=1}^M\}_{k=1}^K$, Weights $W = \{W_k\}_{k=1}^K$
 - 3: **Output:** M -quantile projection of the mixture distribution $\mathcal{P}_Q^M(\eta_{\text{mix}})$, where $\eta_{\text{mix}} = \sum_{k=1}^K W_k \eta_k$
 - 4: Concatenate values $X_{\text{mix}} = (X_1, X_2, \dots, X_K)$
 - 5: Compute new weights for each particle in X_{mix} as the original weight times the weight in W
 - 6: Sort X_{mix} in ascending order as $\bar{X}_{\text{mix}}$
 - 7: Order the associated weights for each particle in $\bar{X}_{\text{mix}}$
 - 8: Find in $\bar{X}_{\text{mix}}$ the value corresponding to each quantile level for a new vector $\tilde{X}_{\text{mix}}$
 - 9: **Return:** $\tilde{X}_{\text{mix}}$
-

3.4 The Implementation

In this subsection, we collect all the discussions so far to describe the workflow for the distributional approach, specifically choosing a particular measure ρ and the M -quantile characterization and projection for the distribution function. Again, the goal is to solve an MDP, and after initializing a value distribution function $\eta : \eta(s, a) \in \mathcal{F}_Q^M, \forall s \in \mathcal{S}, a \in \mathcal{A}$, we iteratively impose the distributional Bellman operator as in Equation 6, together with the M -Quantile Projection.

$$(\mathcal{T}\eta)(s, a) = \mathcal{B}_{r, \gamma} \mathcal{P}_Q^M \left(\int_{s' \in \mathcal{S}} \eta(s', a') f(s' | s, a) ds' \right) \quad (10a)$$

$$\text{where } a' = \arg \max_{a' \in \mathcal{A}} \rho(\eta^*(s', a')) \quad (10b)$$

$$\eta^* = \mathcal{T}\eta^* \quad (10c)$$

We make a concrete statement in Algorithm 2 below.

Algorithm 2 Distributional Dynamic Programming with M -Quantile Projection

- 1: **Parameters:** An MDP tuple $(\mathcal{S}, \mathcal{A}, \gamma, P_{\mathcal{S}}, P_{\mathcal{R}})$; number of quantile levels M and the associated $\tau_M = \left\{ \frac{2m-1}{2M} \right\}_{m=1}^M$; measure ρ ; a metric d and threshold ϵ for convergence checking
 - 2: **Initialization:** The M -quantile projected value distribution function $\eta^0 \in \mathcal{F}_Q^M : \mathcal{S} \times \mathcal{A} \times \tau_M \rightarrow \mathbb{R}$ stored in a $|\mathcal{S}| \times |\mathcal{A}| \times M$ matrix
 - 3: **for** $i = 0$ to $MaxIter$ **do**
 - 4: **Step 1: Maximization over ρ**
 - 5: **for** each $s' \in \mathcal{S}$ **do**
 - 6: Choose $\eta^i(s', a')$ such that $a' = \arg \max_{a' \in \mathcal{A}} \rho(\eta^i(s', a'))$
 - 7: **end for**
 - 8: **Step 2: Integration and Projection**
 - 9: Implement Algorithm 1 with distributions $\{\eta^i(s', a')\}_{s' \in \mathcal{S}}$ and weights $\{f(s'|s, a)\}_{s' \in \mathcal{S}}$
 - 10: **Step 3: Scaling**
 - 11: Apply the bootstrap operator $\mathcal{B}_{r, \gamma}$ to the outcome from Step 2 to obtain η^{i+1} for the next iteration
 - 12: **Check:** If $d(\eta^i, \eta^{i+1}) < \epsilon$, set $\eta^* = \eta^{i+1}$, break, and report convergence
 - 13: **end for**
 - 14: If convergence, return η^* and the associated policy function $a^*(s) = \arg \max_{a \in \mathcal{A}} \rho(\eta^*(s, a))$
-

In the next two sections, we apply the distributional methods to two classical structural models. Specifically, in Section 3, we examine the McCall Search Model in labor economics, exploring how an agent's choice of distribution measure ρ influences their value distribution

function and, consequently, their reservation wage when deciding whether to accept an offer. In Section 4, we turn to the Aiyagari Model in macroeconomics to investigate how the choice of ρ affects the wealth distribution in general equilibrium. For both models, we use the calibration from [Sargent and Stachurski \[2014\]](#), which offers a simplified version that highlights the key features of the models and provides fast source code for the benchmark VFI solution.

4 Case 1: McCall Search Model

The McCall search model [\[McCall, 1970\]](#) is a fundamental framework in labor economics that analyzes the decision-making process of unemployed workers as they search for job offers. This model addresses the trade-off between accepting a current job offer and continuing to search for potentially better opportunities in the future. It is particularly useful for understanding the dynamics of unemployment, job search behavior, and the impact of unemployment benefits on the job market.

4.1 Model Setup and Benchmark Solution

A McCall worker observes a wage offer w_t in each period drawn from $F(\cdot)$ and he wants to maximize an expected discounted sum of lifetime incomes $\mathbb{E} \sum_{t=0}^{\infty} \gamma^t y_t$, where the income y_t equals his wage w_t if he is employed and the constant unemployment compensation c if he is unemployed.

In recursive form, the state space is $\mathcal{S} = [w_{\min}, w_{\max}]$ and the action space is $\mathcal{A} = \{A, R\}$ indicating acceptance or rejection. The Q -function satisfies the following system of equations:

$$Q(w, A) = \frac{w}{1 - \gamma}, \quad (11a)$$

$$Q(w, R) = c + \gamma \int_{w' \in \mathcal{W}} \max_{a' \in \{A, R\}} Q(w', a') dF(w'). \quad (11b)$$

where $\frac{w}{1 - \gamma}$ represents the present value of accepting the wage offer w , and the right-hand side of Equation (11b) represents the value for which the agent receives the unemployment compensation c and in the next period compares acceptance and rejection for each potential offer w' .

For implementation, we take two steps. First we discretize the state space into a set of J number of grid points: $\{w_1, w_2, \dots, w_J\}$, indexed by j , with the corresponding probability mass $\{\mathbb{P}(w_1), \dots, \mathbb{P}(w_J)\}$. Second, we make use of the fact that w_t is i.i.d drawn in each period, so $Q(w, R)$ is the same across all w , denoting as:

$$Q_R := Q(w, R) \quad \forall w \in \mathcal{W}.$$

Since $Q(w, A)$ is already determined by Equation (11a), solving the system of equation (11) breaks down to have the following iteration:

$$Q_R^{i+1} = c + \gamma \sum_{j=1}^J \max\left\{\frac{w'}{1-\gamma}, Q_R^i\right\} \mathbb{P}(w'_j) \quad (12)$$

4.2 The Distributional Approach

Next, we introduce the distributional approach for this problem. Our target is the value distribution function η that satisfies the following system of equations:

$$\eta(w, A) = \delta_{\frac{w}{1-\gamma}} \quad (13a)$$

$$\eta(w, R) = \mathcal{B}_{c,\gamma} \int_{w' \in \mathcal{W}} \eta(w', a') dF(w') \quad (13b)$$

$$\text{where } a' = \arg \max_{a' \in \{A, R\}} \rho(\eta(w', a')) \quad (13c)$$

which states that once the agent chooses to accept the offer w , the return is determined and hence distribute according to a delta function $\delta_{\frac{w}{1-\gamma}}$. On the other hand, the distribution of rejecting the offer w is a weighting of a continuum of distributions indexed by the next offer w' , conditional on which the action a' is chosen according to a selection criterion characterized by the operator ρ between two distributions.

Again we discretize the state space and make use of the fact that $\eta(w, R)$ is invariant across w , denoting as:

$$\eta_R := \eta(w, R) \quad \forall w \in \mathcal{W}.$$

To apply Algorithm 2 we consider the M -quantile projection so that η_R is characterized by M particles associated with the quantile levels τ_M . On the other hand $\eta(w, A)$ is represented by a M -length vector with all elements $\frac{w}{1-\gamma}$. Similarly to the case of the Q -function, the system of equation (13) reduces to the following iteration:

$$\eta_R^{i+1} = \mathcal{B}_{c,\gamma} \mathcal{P}_Q^M \sum_{w' \in \mathcal{W}} \kappa_\rho(\delta_{\frac{w'}{1-\gamma}}, \eta_R^i) \mathbb{P}(w') \quad (14)$$

where κ_ρ is an operation that picks between the two distributions that have the higher

value with respect to the measure ρ .

4.3 Calibration and Results

Table 1 lists the structural parameters and hyperparameters chosen for the benchmark case.

Parameters	Definition	Value
γ	The discount rate	0.99
c	The unemployment compensation	25
$w_{\min}$	The lower bound of state space	10
$w_{\max}$	The upper bound of state space	60
J	Number of states in discretization	11
F	The distribution of wage w	Beta($n = 10, \alpha = 200, \beta = 100$)

Table 1: Parameters of the McCall Search Model

Figure 3 illustrates the wage distribution. The asymmetric distribution has little weight on the lower extreme and a very fat tail.

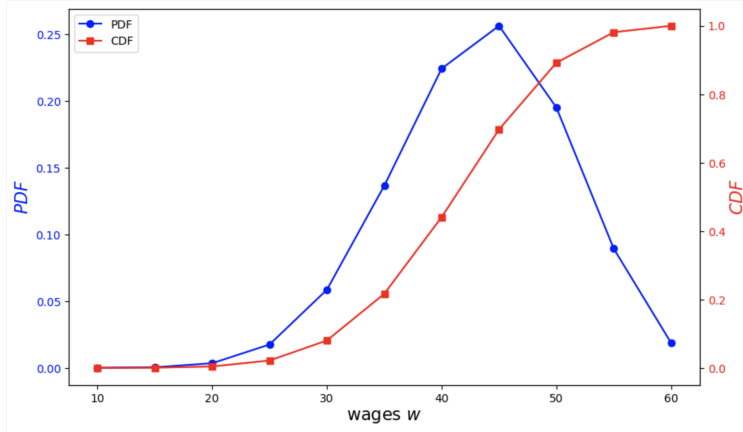


Figure 3: Wage w distribution: BetaBinomial($n = 10, \alpha = 200, \beta = 100$)

Figure 4a shows the benchmark result where we choose $M = 100$. Specifically, the blue line represents η_r in its CDF form. Note that while the theoretical lower bound of the support of η_r is $\frac{w_{\min}}{1-\gamma} = 1000$, determined by accepting the lowest wage offer, Algorithm 2 returns an actual lower bound of $\frac{c}{1-\gamma} = 2500$, determined by the agent's reservation option to reject the offer and receive unemployment compensation indefinitely. This indicates the success of the implementation. On the other hand, Algorithm 2 does return the same upper bound as the theoretical one, $\frac{w_{\max}}{1-\gamma} = 6000$, which is achievable by accepting the highest wage offer.

The curve of η_r exhibits substantial variation compared to the CDF of wages in Figure 3, and the former is much steeper. For instance, the mean wage is approximately 43.3, while the

mean of η_r , interpreted as the expected return if the offer is rejected, is 5228.43, significantly higher than $\frac{43.3}{1-\gamma} = 4330$. In Figure 3, the 20% quantile of the wage offer is around 40, whereas the 20% quantile of η_r is over 5000. This indicates that the mapping from the wage distribution (state variable) to the value function is not affine, which is attributable to the option of rejection in the structural model.

We compare the mean of η_R with Q_R obtained from the value function iteration in 4.1, which are 5228.43 and 5322.27, respectively. The difference is due to the approximation error introduced by the projection in each iteration, but is relatively small compared to the magnitude of the values, with a relative difference of only $\frac{5322.27-5228.43}{5228.43} = 1.7\%$. Figure 4 demonstrates that the approximation error in the value has no significant impact on the derived policy function: both approaches yield the same action for each wage offer.

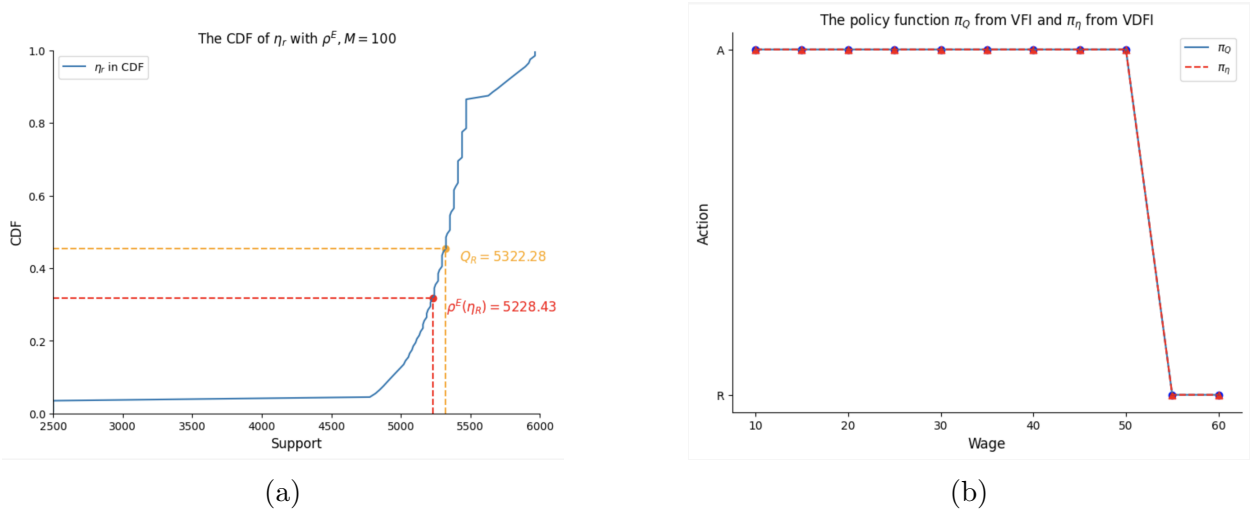


Figure 4: (a) The CDF of η_r and (b) the policy functions π_Q and π_η .

As a robustness check, Figure 5 demonstrates the relative differences between $\rho(\eta_R)$ and Q_R in a range of options for M in $[10, 20, 50, 100, 200, 1000]$. The results suggest that the M -quantile projection may potentially cause an underestimation of the mean of the distribution compared to the VFI result. The underlying reasons for this discrepancy require further investigation, which will be addressed in future extensions of this work. Nevertheless, the difference is substantially small, even for a very low choice of $M = 10$, and the discrepancy narrows as M increases. In this exercise, we choose $M = 100$ as the benchmark, balancing both the accuracy of the distribution approximation and the computational speed.

Solutions for Different Measures ρ Figure 6 illustrates the solution η_R associated with different choices of ρ , where the agent optimizes not the expected value but an alternative measure of the value distribution function.

In Figure 6a, we consider the Mean-Variance measure ρ^{EV} for a range of variance weights: $\lambda = (0.1, 0.5, 1.0)$. The results show that the CDF of η_R becomes more wavy as λ increases, indicating that the agent is more risk averse. Another important aspect is that the range of

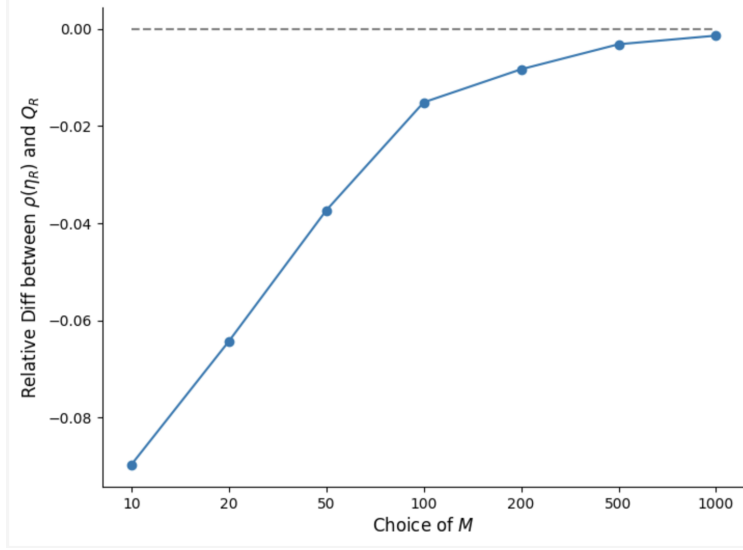


Figure 5: The relative differences between $\rho(\eta_R)$ and Q_R for a range of choices of M

η_R shrinks as λ increases, which is reasonable as it reflects the agent's preference for more certain returns.

In Figure 6b, we examine the Quantile measure ρ^Q across different quantile levels: $\tau = (0.2, 0.4, 0.6, 0.8)$. Given that the wage distribution has a fat tail, the η_R associated with higher τ values is closer to the case where the agent maximizes the expected value. In contrast, the η_R associated with lower values τ has a higher lower bound, which makes sense as the agent seeks a higher value for the lower quantile level.

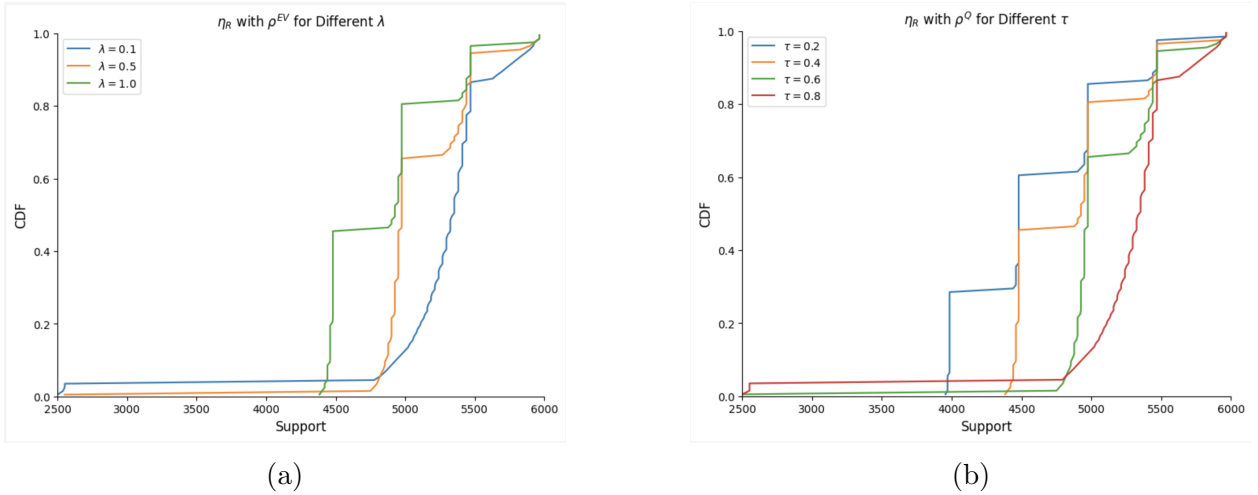


Figure 6: (a) η_r of the Mean-Variance measure ρ^{EV} for different λ ; (b) η_r of the Quantile measure ρ^Q for different τ .

5 Case 2: Aiyagari Model

The Aiyagari model [Aiyagari, 1994] is a foundational framework in macroeconomics that explores the interactions between income inequality, savings behavior, and aggregate economic outcomes. This model extends the standard neoclassical growth model by incorporating heterogeneous agents who face idiosyncratic income shocks and have limited access to financial markets. As a result, it provides valuable insights into the distribution of wealth, the role of precautionary savings, and the effects of incomplete markets on capital accumulation and economic growth. The Aiyagari model is particularly useful for analyzing how individual income risk and borrowing constraints influence aggregate savings and the distribution of wealth in an economy.

5.1 Model Setup

Household

There is a continuum of infinitely lived and ex-ante identical households. In each period, households face a binary productivity shock ϵ_t , which follows an exogenous Markov chain characterized by a transition matrix Π . The households own capital k_t , which they use for both consumption c_t and investment. The households are subject to a non-zero borrowing constraint, $k_t \geq 0$, due to incomplete markets. The optimization problem faced by an agent is:

$$\max \mathbb{E} \sum_{t=0}^{\infty} \gamma^t \log(c_t) \quad (15)$$

subject to

$$k_{t+1} + c_t \leq w\epsilon_t + (1+r)k_t \quad (16)$$

$$k_t \geq 0 \quad (17)$$

where w and r represent the per unit labor productivity and the interest rate, respectively. These two market prices are taken as given from the perspective of households and are determined by aggregation through firms' optimization.

Firms

The goods market is competitive, and firms face constant returns to scale, with production following a Cobb-Douglas form:

$$Y = AK^\alpha L^{1-\alpha} \quad (18)$$

where $A > 0$ and $\alpha \in (0, 1)$ are parameters representing total-factor productivity and capital share, respectively. $K_t = \int k_t dF(k_t)$ denotes the aggregate capital holdings and L represents the aggregate labor supply, which is exogenously determined by Π . The first-order condition of the firms' maximization problem

$$\max_K \{AK^\alpha L^{1-\alpha} - (r + \delta)K - wL\}$$

determines the market prices:

$$r = A\alpha \left(\frac{L}{K}\right)^{1-\alpha} - \delta \quad (19)$$

$$w = A(1 - \alpha) \left(\frac{L}{K}\right)^{-\alpha} \quad (20)$$

We can further combine (19) and (20) to express the wage w as a function of the interest rate r :

$$w(r) = A(1 - \alpha) \left(\frac{A\alpha}{r + \delta}\right)^{\alpha/(1-\alpha)} \quad (21)$$

so that we only need to track r as the price in the system. Note that since there are no aggregate shocks in the economy, the distribution of capital holdings and labor productivities are constant over time in equilibrium. Consequently, all the aggregate variables Y , K , and L , as well as the prices r and w , remain constant, allowing us to drop the time subscript.

Equilibrium

We consider a stationary rational expectations equilibrium (SREE) for this problem:

1. Given the supply price r_s (and consequently w), households solve their savings problem (15) subject to (16) and (17), leading to a constant aggregate supply K .
2. The firm's optimization problem (18) lead to the capital demand function (19).
3. The capital market clears and the demand price is consistent with the supply price:
 $r_s = r_d$.

5.2 The Computation Strategy

The state space is defined as $\mathcal{S} = [k_{\min}, k_{\max}] \times \{\epsilon_{\text{low}}, \epsilon_{\text{high}}\}$, where the state variable is the pair representing both capital holdings and productivity, $s = (k, \epsilon)$. The action space is $\mathcal{A} = [k_{\min}, k_{\max}]$, with the action being the chosen capital holding for the next period, $a = k'$. The feasibility of k' , given the current state (k, ϵ) , is constrained by the budget constraint of the household (16). We enforce this constraint by defining the reward function $R(s, a)$ such that any state-action pair violating (16) results in a reward value of negative infinity.

The computation of SREE involves a two-layer algorithm:

In the second layer, we solve the agent's savings problem for a given price r_s using VFI or DVFI to obtain the agent's optimal policy π^* . We then compute the stationary capital distribution $F(K)$ and, consequently, the aggregate supply K . Next, we determine the market-clearing condition by considering the firm's response to K , which allows us to compute the demand price r_d .

In the first layer, we use a bisection algorithm to find the equilibrium price r^* . Specifically, we begin by proposing two supply prices (r_{s1}, r_{s2}) such that the former is too low (leading to excess demand for K , and hence $r_{d1} > r_{s1}$) and the latter is too high ($r_{d2} < r_{s2}$), ensuring that r^* lies within the interval $[r_{s1}, r_{s2}]$. We then consider the midpoint of the interval $r_{s3} = \frac{r_{s1} + r_{s2}}{2}$, and replace r_{s1} or r_{s2} with r_{s3} depending on whether r_{s3} is too low or too high, thus narrowing the interval. This process is repeated until the lower and upper bounds of the interval converge closely.

5.3 Calibration and Results

Table 2 lists the structural parameters and hyperparameters chosen for the benchmark case. To discretize the continuous space $\mathcal{S} = [k_{\min}, k_{\max}]$ we choose J grid points $(k_1, \dots, k_J)$, following the polynomial rule [Maliar and Maliar, 2014]:

$$k_j = \left(\frac{j}{J}\right)^d (k_{\max} - k_{\min}) + k_{\min} \quad \text{for } j = 1, \dots, J \quad (22)$$

which enable more grids at the low capital levels and help in accurate approximation. The number of grids in the discretization J is chosen to balance computational efficiency with the need to sufficiently capture the features of the model.

Next, we consider the results for different choices of M in the M -quantile parameterization of the distribution, as illustrated in Figure 7. Since visualizing the Q -function and the η -function in this context is challenging, we instead focus on two one-dimensional functions of capital k for illustration.

On the left-hand side of Figure 7, we plot the value functions $V(k, \epsilon)$ obtained after

Parameters	Definition	Value
γ	The discount rate	0.96
α	The capital share	0.33
A	TFP	1
L	Aggregate labor supply	1
$(k_{\min}, k_{\max})$	The lower and upper bound of capital	(0, 10)
$(\epsilon_{\text{low}}, \epsilon_{\text{high}})$	The binary productivity	(0.1, 1)
Π	The Markov Chain characterizing ϵ	$\begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$
d	Degree of the polynomial rule discretization	2
J	Number of grid in discretization	20

Table 2: Parameters of the Aiyagari Model

solving for Q and η , for better visualization. We compare the cases of VFI and DVFI with different M values against the capital k on the x-axis, each with two lines corresponding to ϵ_{low} (lower curve) and ϵ_{high} (upper curve). Similarly to the results in Case 1, higher M values provide a better approximation to the VFI solution, and specifically, when $M = 200$, the difference is almost indistinguishable. On the right-hand side, we plot the stationary distribution function of capital k , characterizing the general equilibrium of the economy obtained from the agents’ optimal policy. The distribution is degenerated and concentrated on a mass when M is small. In particular, when M is sufficiently high, we approximate the case where there is a mass of hand-to-mouth agents at the lower bound of capital, a key feature of the Aiyagari economy. The distribution matches closely that obtained in the VFI case when $M = 200$.

6 Discussion on Possible Extensions

Aside from the topics we have already discussed in this paper, there are still many areas for further exploration in future drafts or related projects. Below, we outline a few key points that we believe are important and interesting:

1. **Parameterized Functional Forms of Distributions:** While this paper focuses on M -quantile parameterization, representing a distribution with M equally weighted particles at particular quantile levels, other distribution representations might be more advantageous depending on the context and could potentially improve computational efficiency. For example, one possible approach is to adopt a specific functional form for the distribution and update the parameters either analytically or through incremental methods like gradient descent. However, a challenge is that many such parameterizations are not closed under the distributional Bellman operator. For instance, param-

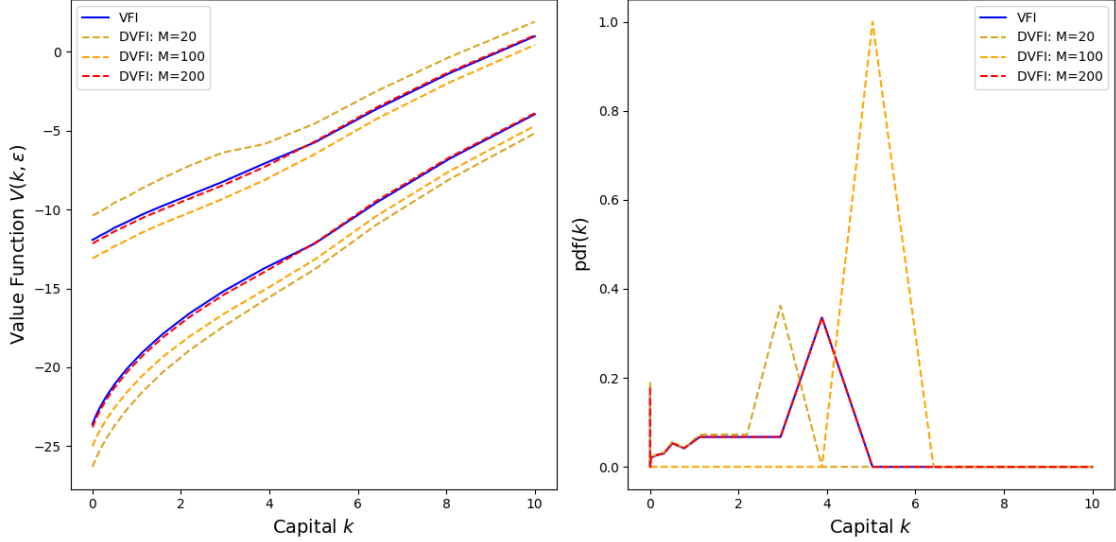


Figure 7: Performance for different choices of M (20, 100, 200). Left: The value functions $V(k, \epsilon)$, each with two lines corresponding to ϵ_{low} (lower curve) and ϵ_{high} (upper curve). Higher values of M provide a better approximation of the correct VFI solution, with $M = 200$ yielding results very close to the exact solution. Right: The asset distribution, $\text{pdf}(k)$. The distribution is more degenerate for smaller M , while for $M = 200$, it closely matches the distribution obtained in the VFI case.

eterizing the distribution as normal would result in a mixture of normal distributions after applying the distributional Bellman operator in one iteration. This necessitates a degree of flexibility in the functional form. Neural networks could be a viable option, as there has been growing literature in economics on parameterizing the value function or policy function using neural networks (see, for example, [Azinovic et al. \[2022\]](#), [Maliar et al. \[2021\]](#), [Maliar and Maliar \[2022\]](#), [Gu et al. \[2023\]](#), [Han and Yang \[2021\]](#), [Kahou et al. \[2021\]](#)). Parameterizing the value distribution function with a neural network could fall within this domain.

2. **Computational Complexity:** In this paper, we have not extensively discussed the computational complexity and implementation effectiveness of the methods presented, mainly because our focus was on the new economic insights provided by the distributional approach, and the examples we used involve classical models that can be solved quickly, even when extended to the distributional approach. Ideally, the computational complexity of the distributional methods compared to the traditional value function approach would be $\mathcal{O}(M)$, as the transition is from tracking a scalar for each state-action pair to now tracking a distribution represented by M particles. However, this depends on the specific algorithm's implementation, and the complexity could change if we consider alternative ways of parameterizing the distribution function, as discussed in the previous point.

3. **Economic Interpretation:** In this paper, we have considered two classical structural models, the McCall search model and the Aiyagari model, both for their established economic insights and because mature routines exist for solving these models using traditional value function approaches, allowing for easy comparison with the distributional approach. Whether the distributional approach can provide enough new insights to justify the additional complexity and slower computation depends on the context. As future extensions, we aim to identify other structural models that might be well-suited to the application of our distributional methods.

7 Conclusion

This paper has introduced a novel framework for decision-making in structural economic environments by focusing on the entire return distribution rather than just the expected return. By extending the traditional expectation maximization approach to include the maximization of various measures over the distribution function, we have provided a more comprehensive understanding of agent behavior under uncertainty. This approach allows for a broader analysis, incorporating risk preferences and other factors that are typically overlooked when focusing solely on the first moment of the distribution.

We applied our framework to two foundational models in economics: the McCall Search Model in labor economics and the Aiyagari Model in macroeconomics. In these applications, we demonstrated how considering the full distribution of outcomes, rather than just the expected value, can lead to different optimal policies and equilibrium outcomes. The McCall Search Model highlighted how risk aversion and concern for specific quantiles can influence the decision to accept or reject job offers. The Aiyagari model showed how different measures of the value distribution function can affect the stationary distribution of wealth in an economy.

Our proposed *Distributional Value Function Iteration (DVFI)* method offers a powerful tool to analyze dynamic decision-making processes in economics. Using the *distribution Bellman operator* and the *M*-quantile parameterization, we have provided a practical and theoretically sound approach to solving these complex problems. This method not only generalizes traditional dynamic programming, but also opens new avenues for research in areas where agents' preferences over distributional outcomes play a critical role.

Future research could explore further applications of this framework to other economic models, investigate alternative parameterizations of the value distribution function, and examine the implications of using different measures on policy outcomes. The potential to better capture the complexity of human decision-making under uncertainty makes this a promising direction for ongoing and future work in economics.

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